

DHCP – Packet Tracer Activity

In this activity you will create a simple network and configure a router to act as a DHCP server. Once configured, your hosts will be configured to automatically obtain IP addresses from the DHCP server.

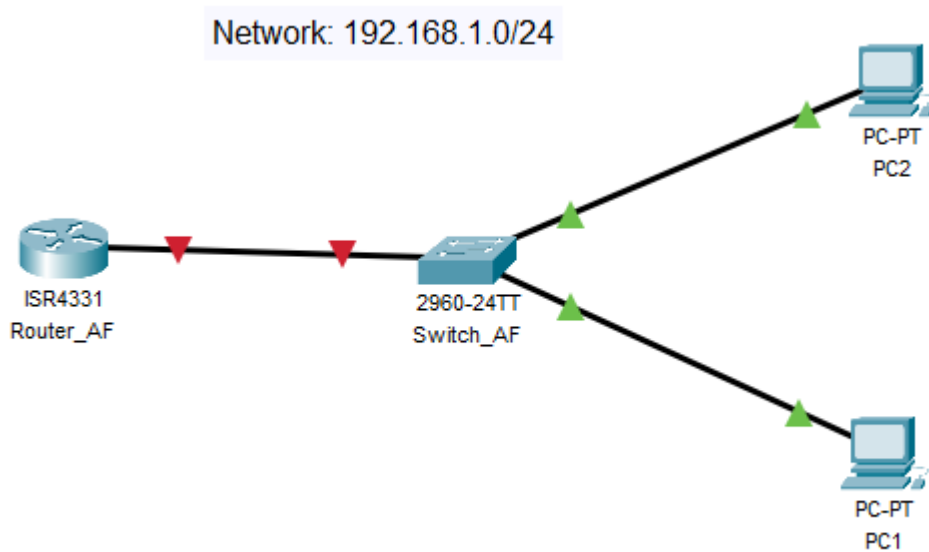
Submission requirements

Follow the steps provided in this activity to create a topology and configure DHCP server and clients. Document the step-by-step process together with the appropriate screenshots while you are doing the configuration. Submit the report to your trainer.

Step 1: Network topology

Open the packet tracer and create the following topology. The router and switch version could be different what the diagram given below.

Rename the Router and the switch to include your initials.



Step 2: Activate the router interface

Double click on the router to open the router CLI.

Enter 'no' for the prompt 'Would you like to enter the initial configuration dialog? [yes/no]:'

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

% Please answer 'yes' or 'no'.

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

```
Router>  
Router>
```

Activate the router interface you used to connect (Eg. G0/0/0). Actual interface you used in your topology may vary.

1. Go to the interface using **interface *interface_id*** command
2. Assign an ip address to the interface using **ip address *ip_address subnet_mask*** command
3. Activate the interface using **no shutdown** command

Step 3: Exclude IP addresses

Following IP address range should be excluded from the dhcp pool as they are planned to be assigned as the IP addresses for intermediate network devices in the organization.

IPs to exclude: 192.168.1.3 – 192.168.1.13

Step 4: Define a DHCP pool name.

Use ***YourName_Pool*** as the pool name. (Replace **YourName** with your real name)

Step 5: Configure DHCP Pool

1. Set the network as 192.168.1.0/24
2. Set the default-router IP as 192.168.1.3

Step 6: Configure hosts to obtain ips from DHCP server

Double click on PC and go to 'IP Configuration'.

Select 'DHCP' radio button.

Check and verify a correct IP is assigned through DHCP